

$$\int dx = x + c$$

$$\begin{aligned} & \int 2 \cdot dx \\ &= 2 \cdot \int dx \\ &= 2x + c \end{aligned}$$

Integral de x elevado a la n

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$

$$\int x^3 dx$$

$$= \frac{x^{3+1}}{3+1} + c$$

$$= \frac{x^4}{4}$$

$$\int x^7 dx = \frac{x^8}{8} + c$$

$$\int t^4 dt = \frac{t^5}{5} + c$$

$$\int x^{-3} dx = \frac{x^{-2}}{-2} + c = -\frac{x^{-2}}{2} + c$$

$$\int x^{-1} dx = \ln x + c$$

$$\int x^5 dx$$

$$= \frac{x^6}{6} + c$$

$$\int x^n dx = \frac{x^{n+1}}{n+1}$$

$$\int \frac{3}{x^3} dx$$

$$= \int 3x^{-3} dx$$

$$= 3 \cdot \int x^{-3} dx$$

$$= 3 \cdot \frac{x^{-3+1}}{-3+1} + C = \frac{3x^{-2}}{-2} + C$$

Integral de un polinomio

$$\int (2x^3 + 5x^2 - 2x + 1) dx$$

$$= \frac{x^4}{2} + \frac{5x^3}{3} - x^2 + x + C$$

Integral de un polinomio

$$\begin{aligned}& \int \frac{6x^5 - 12x^4 + 9x^3}{3x^2} dx \\&= \int \left(\frac{\overset{2}{\cancel{6}}x^5}{\underset{1}{\cancel{3}}x^2} - \frac{\overset{4}{\cancel{12}}x^4}{\underset{1}{\cancel{3}}x^2} + \frac{\overset{3}{\cancel{9}}x^3}{\underset{1}{\cancel{3}}x^2} \right) dx \\&= \int (2x^3 - 4x^2 + 3x) dx \\&= 2 \int x^3 dx - 4 \int x^2 dx + 3 \int x dx\end{aligned}$$

$$\begin{aligned}&= \cancel{2} \cdot \frac{x^4}{\underset{2}{\cancel{4}}} - 4 \cdot \frac{x^3}{3} + 3 \cdot \frac{x^2}{2} + C \\&= \frac{x^4}{2} - \frac{4x^3}{3} + \frac{3x^2}{2} + C\end{aligned}$$

Integral de un polinomio

$$\begin{aligned} & \int \frac{5x^3 - 3x^2 + 2}{2x^2} dx \quad \frac{a^2}{b^3} \\ &= \int \left(\frac{5x^3}{2x^2} - \frac{3x^2}{2x^2} + \frac{2}{x^2} \right) dx \\ &= \int \frac{5x}{2} dx - \int \frac{3}{2} dx + \int \frac{1}{x^2} dx \\ &= \frac{5}{2} \int x dx - \frac{3}{2} \int dx + \int 1x^{-2} dx \end{aligned}$$

$$\begin{aligned} & \int x^n dx = \\ &= \frac{5}{2} \cdot \frac{x^2}{2} - \frac{3}{2} x + \frac{1x^{-1}}{-1} + C \\ &= \frac{5x^2}{4} - \frac{3x}{2} - \frac{1}{1x} \end{aligned}$$

Integral de un polinomio

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C$$

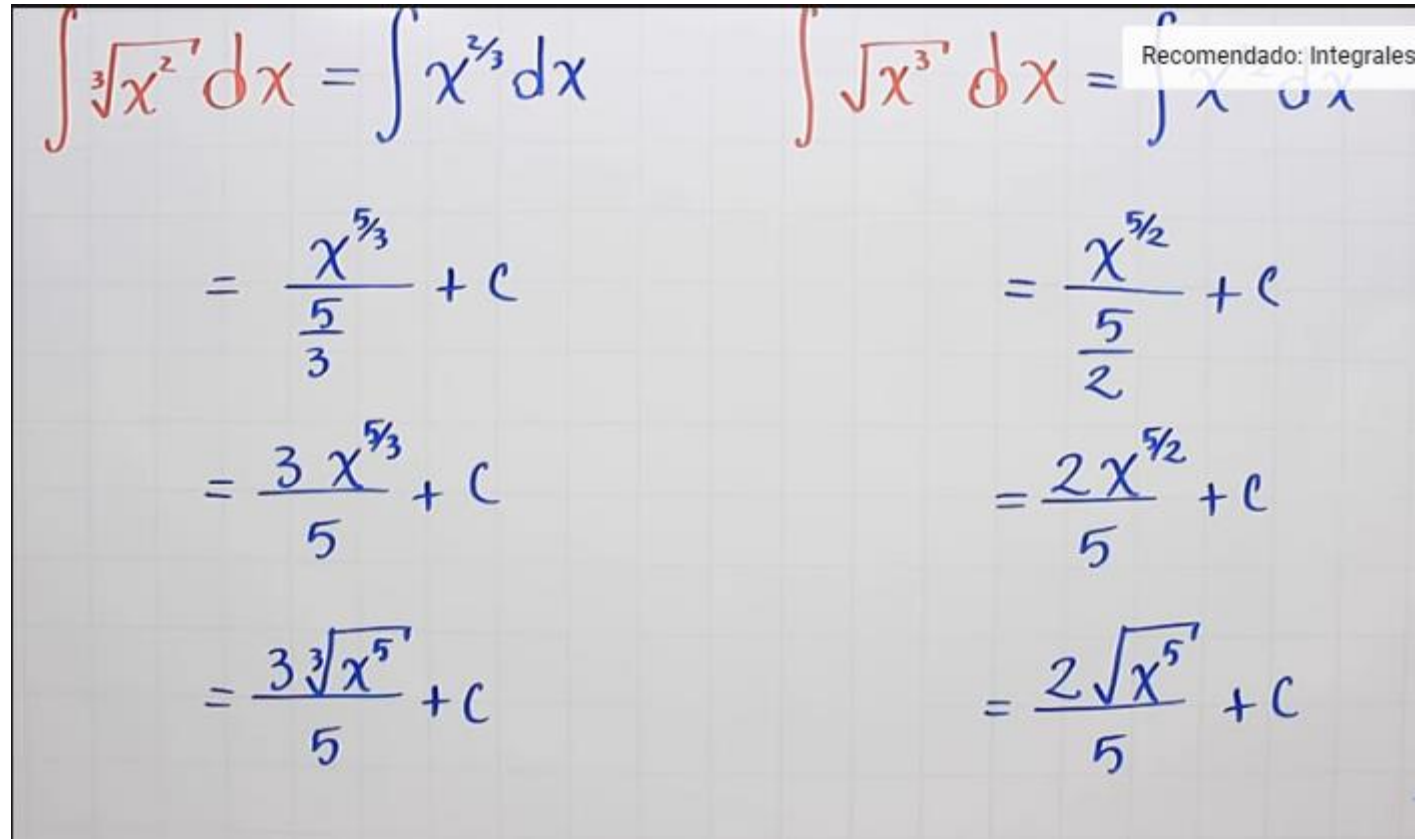
$$\int \sqrt{x} dx = \int x^{1/2} dx$$

$$= \frac{x^{1/2+1}}{1/2+1} + C$$

$$= \frac{x^{3/2}}{\frac{3}{2}} + C$$

$$= \frac{2x^{3/2}}{3} + C$$

Integral de un polinomio



The image shows two handwritten mathematical derivations on a grid background. The first derivation on the left calculates the integral of the cube root of x squared, showing the conversion to a power of 2/3, the application of the power rule to get x^(5/3) divided by 5/3, and the final result as (3/5) times the cube root of x to the power of 5. The second derivation on the right calculates the integral of the square root of x cubed, showing the conversion to a power of 3/2, the application of the power rule to get x^(5/2) divided by 5/2, and the final result as (2/5) times the square root of x to the power of 5. A small text box in the upper right corner of the grid area reads 'Recomendado: Integrales'.

$$\int \sqrt[3]{x^2} dx = \int x^{2/3} dx$$
$$= \frac{x^{5/3}}{\frac{5}{3}} + C$$
$$= \frac{3x^{5/3}}{5} + C$$
$$= \frac{3\sqrt[3]{x^5}}{5} + C$$
$$\int \sqrt{x^3} dx = \int x^{3/2} dx$$
$$= \frac{x^{5/2}}{\frac{5}{2}} + C$$
$$= \frac{2x^{5/2}}{5} + C$$
$$= \frac{2\sqrt{x^5}}{5} + C$$

Recomendado: Integrales

Integral de un polinomio

$$\int x^n dx = \frac{x^{n+1}}{n+1} + c$$
$$\sqrt[m]{x^n} = x^{n/m}$$
$$\int \sqrt{3x} dx$$
$$= \int \sqrt{3} \cdot \sqrt{x} dx$$
$$= \sqrt{3} \cdot \int \sqrt{x} dx$$

Integral de un polinomio

$$\int \frac{5}{\sqrt{x}} dx$$
$$= \int \frac{5 dx}{x^{1/2}}$$

$$\sqrt[m]{x^n} = x^{n/m}$$

$$\frac{x}{y^n} = x \cdot y^{-n}$$

Integral de una multiplicación

$$\begin{aligned} & \int 3x^2(2x^2 - 5) dx & \int x^n \\ &= \int (6x^4 - 15x^2) dx \\ &= \int 6x^4 dx - \int 15x^2 dx \\ &= 6 \int x^4 dx - 15 \int x^2 dx \end{aligned}$$

$$\int (\underline{3x} - \underline{5})^2 dx$$

$$= \int ((3x)^2 - 2 \cdot 3x \cdot 5 + 5^2) dx$$

$$= \int (9x^2 - 30x + 25) dx$$

$$= \int 9x^2 dx - \int 30x dx + \int 25 dx$$